AI/ML Internship Project

# SPAM EMAIL DETECTION

Aparna Priyadarsini Moharana

POWERED BY MACHINE LEARNING

# INTRODUCTION

## What is Spam?

Spam emails are unsolicited, unwanted messages sent in bulk. They are not just annoying but pose significant security risks to users globally.

- Phishing & Scams
- Malware Distribution
- Security Vulnerabilities

# PROBLEM STATEMENT

## The Digital Overload

Millions of spam emails are dispatched daily across the internet. Manual filtering is no longer a viable option due to the sheer volume and increasing complexity of these messages.

Automating classification using Machine Learning is essential to ensure user privacy and maintain email server efficiency.

# PROJECT OBJECTIVES

### Automation

Build an intelligent system that automatically detects and segregates spam messages.

### Security

Enhance the overall digital communication security by filtering malicious content.

### Optimization

Develop and evaluate a high-accuracy ML model using the SMS Spam Collection Dataset.

# DATASET DESCRIPTION

| Attribute | Details |
| --- | --- |
| Source | SMS Spam Collection (Kaggle/UCI Repository) |
| Total Messages | 5,572 |
| Ham (Legitimate) | 4,825 |
| Spam | 747 |
| Columns | Label (Ham/Spam), Text Message |

*Note: The dataset exhibits a significant class imbalance, which is typical for real-world spam detection scenarios.*

# PREPROCESSING WORKFLOW

### Cleaning

Removal of punctuation, numbers, and stopwords.

### Tokenization

Breaking sentences into individual words or tokens.

### Encoding

Converting textual labels into numerical binaries (0 and 1).

### TF-IDF

Extracting meaningful features from text vectors.

# CLASS DISTRIBUTION

Ham Messages (86.6%)

Spam Messages (13.4%)

Insights: The distribution confirms that spam is the minority class, requiring robust probabilistic modeling.

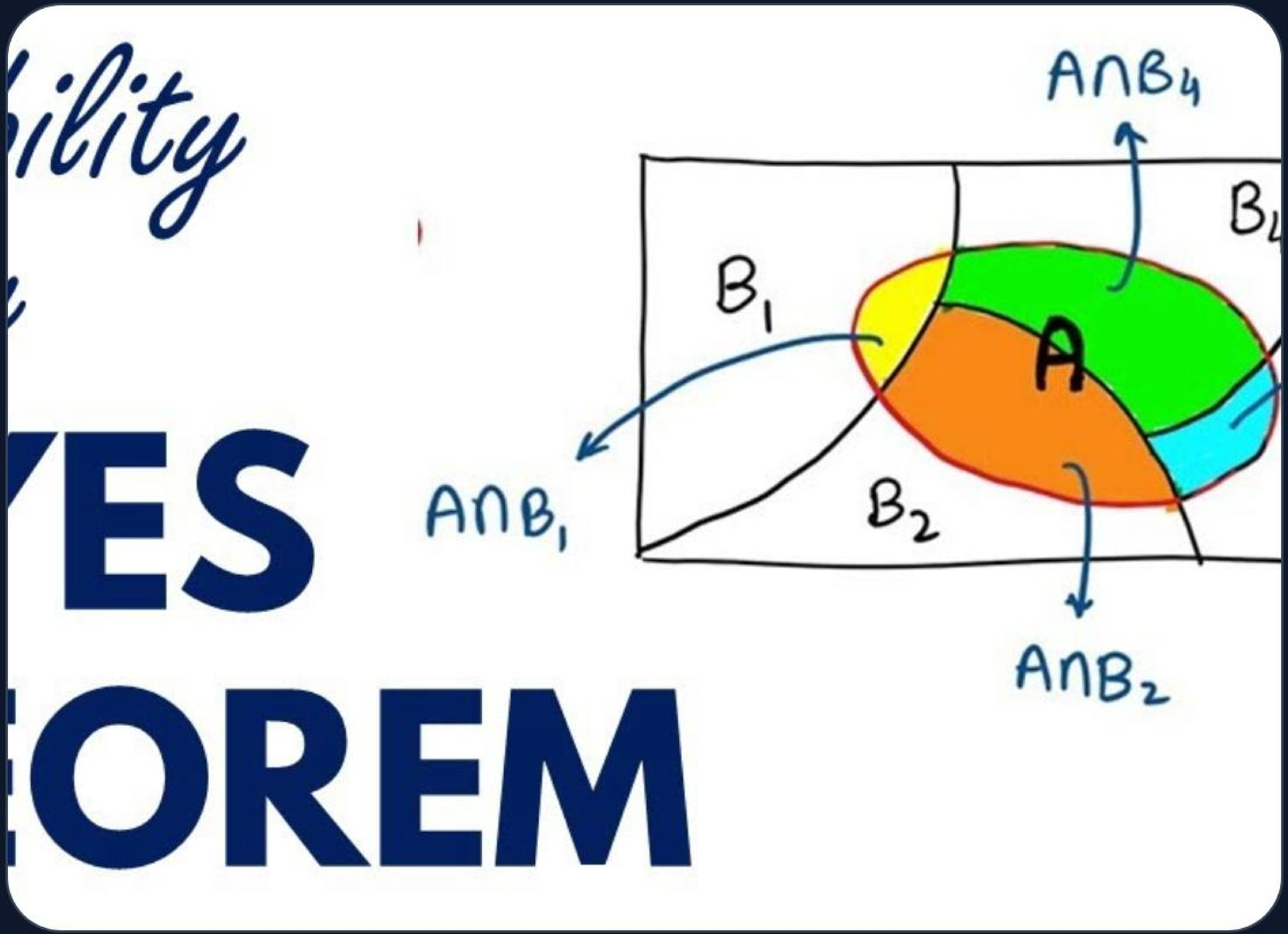
# THE ML MODEL

## Multinomial Naive Bayes

A probabilistic classifier based on Bayes' Theorem with the "naive" assumption of independence between every pair of features.

$$Pc \mid x = \frac{Px \mid cPc}{Px}$$

Ideal for discrete features like word counts in text classification tasks.



# PERFORMANCE METRICS

# 97.6%

## TESTING ACCURACY ACHIEVED

The model demonstrates exceptional reliability in identifying spam while maintaining zero false positives for legitimate (ham) messages.

# CLASSIFICATION REPORT


The Multinomial Naive Bayes model outperforms standard baseline models in both speed and predictive power for text data.

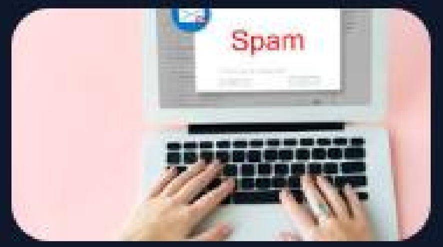
# CONCLUSION & SCOPE

- **Successful Implementation:** Highly accurate spam detection achieved using standard ML techniques.
- **Real-time Deployment:** Integration with email servers for live filtering and message segregation.
- **Deep Learning:** Future exploration using LSTM and Transformers (BERT) for contextual analysis.
- **Multi-language Support:** Expanding the model to detect spam in various global languages.

# THANK YOU

Any Questions?

# IMAGE SOURCES

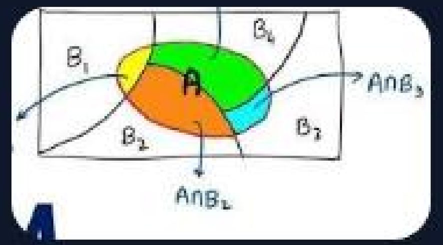



https://cdn.shortpixel.ai/spai/q_lossy+w_949+to_webp+ret_img/mailfloss.com/storage/2024/07/e-mail-popup-warning-window-concept-1024x683.jpg

Source: mailfloss.com


https://png.pngtree.com/thumb_back/fh260/background/20220626/pngtree-information-concept-data-overload-on-digital-background-communication-word-background-photo-image_24913540.jpg

Source: pngtree.com



https://mathematicalmysteries.org/wp-content/uploads/2022/03/bayes-theorem-pt.jpg

Source: mathematicalmysteries.org